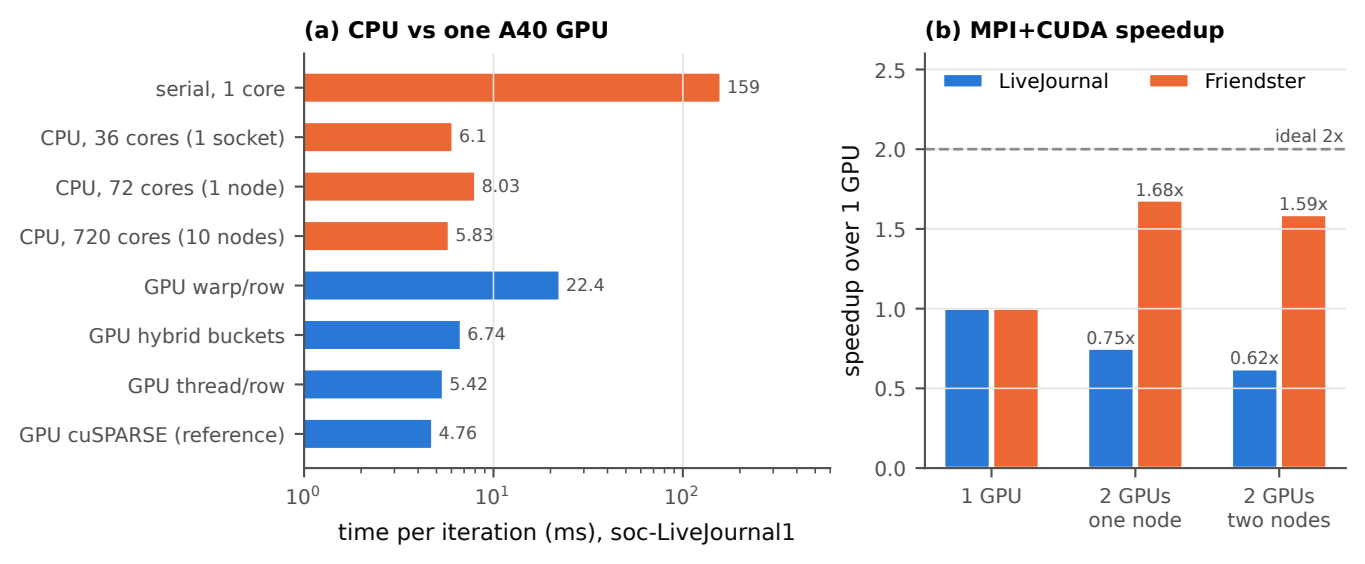
(a) CPU vs one A40 GPU
serial, 1 core
CPU, 36 cores (1 socket)
CPU, 72 cores (1 node)
CPU, 720 cores (10 nodes)
GPU warp/row
GPU hybrid buckets
GPU thread/row
GPU cuSPARSE (reference)
159
6.1
8.03
5.83
22.4
6.74
5.42
4.76
10^0
10^1
10^2
time per iteration (ms), soc-LiveJournal1
(b) MPI+CUDA speedup
LiveJournal
Friendster
ideal 2x
speedup over 1 GPU
2.5
2.0
1.5
1.0
0.5
0.0
1.68x
1.59x
0.75x
0.62x
1 GPU
2 GPUs one node
2 GPUs two nodes